JET-BORNE SU





INDIA

Total number of fighter jets
600+

Sukhoi Su-30MKI
No: **248**
Russian origin multirole fighter manufactured in India under licence

Dassault Rafale
No: **36**
French 4.5-generation multirole fighter, the most advanced jet IAF has

Mirage 2000
No: **45**
French multirole 4th generation fighter jet

Tejas LCA Mk 1
No: **32**
Indigenous supersonic multi-role fighter

MiG-29
No: **65**
Multirole Russian jet, upgraded to Mig-29 UPG

SEPECAT Jaguar
No: **130**
British-made jet ideal for air-support and ground attack

MiG-21 Bison
No: **40**
Soviet-era workhorse of IAF, currently being phased out

In development:
LCA Mk 1A, Mk 2, AMCA

DELAYS AND CHALLENGES

The shortage of fighter aircraft was identified as far back as 2001, and a proposal for 126 new aircraft initiated. The Medium Multi-Role Combat Aircraft (MMRCA) tender in 2008 concluded in 2012 with the Rafale emerging as the frontrunner among six contenders. However, complexities in the Make in India component in the negotiations between Dassault Aviation and Hindustan Aeronautics Limited (HAL) became a bottleneck and, eventually, the programme was shut down. In 2016, a government-to-government agreement was signed for just 36 Rafales. The off-the-shelf acquisitions marked the only foreign fighter induction since the Sukhoi Su-30MKI entered service in the early 2000s.

It was expected that the indigenous Light Combat Aircraft (LCA) Tejas Mk1 would be speedily inducted, but that too was mired in delays. It was inducted in 2015—14 years after its maiden flight. Today, the IAF operates just two LCA Mk1 squadrons, with the Mk1A variant facing a year-long delay in its 2024 delivery schedule. The pace of delivery by state-owned aeronautical organisations is a cause of concern. As the IAF chief said recently, "Technology delayed is technology denied."

The grim scenario appears worse when one contrasts it with the People's Liberation Army Air Force's (PLAAF) fighter jet force. Pakistan's acquisition of the J-35 is a reminder of how the balance of power is shifting in South Asia. Experts point out that as the region braces for a seismic transformation in aerial warfare, India's challenges are mounting. Air Marshal Anil Chopra (retd), former Director-General of the Centre for Air Power Studies, says. "A national task force is the need of the hour, with a deadline and freedom to pick its team, for fighter aircraft development. The way our adversaries are ramping up their air power, India has no option but to put all its efforts on full throttle before it's too late," he says.

HAL, the manufacturer of such indigenous platforms as the Tejas LCA Mk1 and Mk1A, blames supply chain disruptions at US fighter jet engine manufacturer GE Aerospace and pending certification for new systems for the delays. However, HAL will start production soon at its Nashik division, which will join its two existing production lines in Bengaluru, with each line producing eight LCA Mk1A aircraft annually. This will increase HAL's combined production capacity to 24 jets a year. "Once the Nasik line is functional, it will produce eight aircraft annually. We are planning to ramp up in all three lines to cover up the backlog," says a top HAL official.

According to plans, the HAL Tejas Mark 2,Medium Weight Fighter (MWF), a 4.5 generation machine closer to the Rafale in capabilities, will have its first flight in 2026. Mass production is planned for 2029. But these are very optimistic projections, considering that India is dependent on foreign subsystems and, crucially, jet engines, like GE's F404-IN20 engines for LCA Mk1A. "Setting timelines based on assurances from a foreign OEM (original equipment manufacturer) is truly a bit hypothetical," says a senior military planner, betraying deep scepticism.

The development of India's fifth generation fighter, the AMCA, is crucial to countering the PLAAF. The plan is to develop two variants, with the Mark 2 version featuring a more powerful engine and advanced technologies. In 2009, the Union government had allocated Rs 90 crore, followed by an additional Rs 447 crore, for a feasibility study. After years of delay, the AMCA received the Union